

FIPhO 2026 Syllabus

A. Theoretical Part

Item	Syllabus content	Scope and remarks
1	Mechanics	
(a)	Foundations of the kinematics of a point mass.	Vector description of the position of a point mass, with velocity and acceleration represented as vectors.
(b)	Newton's laws; inertial systems.	Problems may be set on changing mass.
(c)	Closed and open systems; momentum and energy; work and power.	
(d)	Conservation of energy; conservation of linear momentum; impulse.	
(e)	Elastic forces; frictional forces; the law of gravitation; potential energy and work in a gravitational field.	Hooke's law; coefficient of friction ($F/R = \text{const}$); static and kinetic frictional forces; choice of the zero of potential energy.
(f)	Centripetal acceleration; Kepler's laws.	
2	Mechanics of Rigid Bodies	
(a)	Statics; center of mass; torque.	Couples; conditions of equilibrium of bodies.
(b)	Motion of rigid bodies; translation; rotation; angular velocity; angular acceleration; conservation of angular momentum.	Conservation of angular momentum about a fixed axis only.
(c)	External and internal forces; equation of motion of a rigid body around a fixed axis; moment of inertia; kinetic energy of a rotating body.	Parallel-axes theorem (Steiner's theorem); additivity of the moment of inertia.
(d)	Accelerated reference systems; inertial forces.	Knowledge of the Coriolis force formula is not required.
3	Hydromechanics	
	Elementary concepts of pressure, buoyancy, and the continuity law.	No specific questions will be set on this topic, but students are expected to know these elementary concepts.
4	Thermodynamics and Molecular Physics	
(a)	Internal energy; work and heat; the first and second laws of thermodynamics.	Thermal equilibrium; quantities depending on state and quantities depending on process.
(b)	Model of a perfect gas; pressure and molecular kinetic energy; Avogadro's number; equation of state of a perfect gas; absolute temperature.	A molecular approach to simple phenomena in liquids and solids, such as boiling and melting, is also included.
(c)	Work done by an expanding gas, limited to isothermal and adiabatic processes.	Proof of the equation of the adiabatic process is not required.
(d)	The Carnot cycle; thermodynamic efficiency; reversible and irreversible processes; entropy (statistical approach); Boltzmann factor.	Entropy as a path-independent function; entropy changes and reversibility; quasistatic processes.

A. Theoretical Part - continued

Item	Syllabus content	Scope and remarks
5	Oscillations and Waves	
(a)	Harmonic oscillations; equation of harmonic oscillation.	Solution of the equation for harmonic motion; attenuation and resonance, qualitatively.
(b)	Harmonic waves; propagation of waves; transverse and longitudinal waves; linear polarization; the classical Doppler effect; sound waves.	Displacement in a progressive wave and understanding its graphical representation; measurements of the velocity of sound and light; Doppler effect in one dimension only; propagation of waves in homogeneous and isotropic media; reflection and refraction; Fermat's principle.
(c)	Superposition of harmonic waves; coherent waves; interference; beats; standing waves.	Understanding that wave intensity is proportional to the square of its amplitude. Fourier analysis is not required, but candidates should understand that complex waves can be formed by adding simple sinusoidal waves of different frequencies. Interference due to thin films and other simple systems (final formulae are not required); superposition of waves from secondary sources (diffraction).
6	Electric Charge and Electric Field	
(a)	Conservation of charge; Coulomb's law.	
(b)	Electric field; potential; Gauss' law.	Gauss' law is confined to simple symmetric systems such as a sphere, cylinder, or plate; electric dipole moment.
(c)	Capacitors; capacitance; dielectric constant; energy density of an electric field.	
7	Current and Magnetic Field	
(a)	Current; resistance; internal resistance of a source; Ohm's law; Kirchhoff's laws; work and power of direct and alternating currents; Joule's law.	Simple cases of circuits containing non-ohmic devices with known V-I characteristics.
(b)	Magnetic field (B) of a current; current in a magnetic field; Lorentz force.	Particles in a magnetic field; simple applications such as the cyclotron; magnetic dipole moment.
(c)	Ampere's law.	Magnetic fields of simple symmetric systems such as a straight wire, circular loop, and long solenoid.
(d)	Law of electromagnetic induction; magnetic flux; Lenz's law; self-induction; inductance; permeability; energy density of a magnetic field.	
(e)	Alternating current; resistors, inductors, and capacitors in AC circuits; voltage and current resonances (parallel and series).	Simple AC circuits; time constants. Final formulae for the parameters of specific resonance circuits are not required.

A. Theoretical Part - continued

Item	Syllabus content	Scope and remarks
8	Geometric optics	
(a)	Propagation of light	
(b)	Reflection	
(c)	Refraction	
(d)	Mirrors and lenses	
(e)	Optical instruments and devices	
9	Electromagnetic Waves	
(a)	Oscillatory circuit; frequency of oscillations; generation by feedback and resonance.	
(b)	Wave optics; diffraction from one and two slits; diffraction grating; resolving power of a grating; Bragg reflection.	
(c)	Dispersion and diffraction spectra; line spectra of gases.	
(d)	Electromagnetic waves as transverse waves; polarization by reflection; polarizers.	Superposition of polarized waves.
(e)	Resolving power of imaging systems.	
(f)	Black body; Stefan-Boltzmann law.	Planck's formula is not required.
10	Quantum Physics	
(a)	Photoelectric effect; energy and momentum of the photon.	Einstein's formula is required.
(b)	De Broglie wavelength; Heisenberg's uncertainty principle.	
11	Relativity	
(a)	Principle of relativity; addition of velocities; relativistic Doppler effect.	
(b)	Relativistic equation of motion; momentum; energy; relation between energy and mass; conservation of energy and momentum.	
12	Matter	
(a)	Simple applications of the Bragg equation.	
(b)	Energy levels of atoms and molecules (qualitatively); emission and absorption; spectrum of hydrogen-like atoms.	
(c)	Energy levels of nuclei (qualitatively); alpha, beta, and gamma decays; absorption of radiation; half-life and exponential decay; components of nuclei; mass defect; nuclear reactions.	

B. Practical Part

The Theoretical Part of the syllabus provides the basis for all experimental problems. The problems given in the experimental contest should include measurements.

Additional requirements

No.	Requirement
1	Candidates must be aware that measuring instruments can affect the measurements being made.
2	Knowledge of the most common experimental techniques for measuring the physical quantities mentioned in Part A.
3	Knowledge of commonly used simple laboratory instruments and devices, such as calipers, thermometers, simple voltmeters, ohmmeters, ammeters, potentiometers, diodes, transistors, and simple optical devices.
4	Ability to use, with proper instructions, more sophisticated instruments and devices, such as a double-beam oscilloscope, counter, ratemeter, signal and function generators, an analog-to-digital converter connected to a computer, amplifier, integrator, differentiator, power supply, and universal analog and digital voltmeters, ohmmeters, and ammeters.
5	Proper identification of error sources and estimation of their influence on the final result or results.
6	Absolute and relative errors; accuracy of measuring instruments; error of a single measurement; error of a series of measurements; error of a quantity given as a function of measured quantities.
7	Transformation of a dependence into linear form through an appropriate choice of variables, and fitting a straight line to experimental points.
8	Proper use of graph paper with different scales, including polar and logarithmic graph paper.
9	Correct rounding and expression of final results and errors using the appropriate number of significant digits.
10	Standard knowledge of laboratory safety. If an experimental setup contains any safety hazards, appropriate warnings must be included in the problem text.